

Example 2.4: 1. The subset $A_1 = \{-2, 0.5, 7\}$ is bounded. Any number $a \leq -2$ is a lower bound and any number $b \geq 7$ is an upper bound.

2. The subset $A_2 = \mathbb{N}$ is bounded from below but not from above. Any number $a \leq 0$ is a lower bound.

3. The subset $A_3 = \mathbb{Z}$ is unbounded.

4. The subset $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\} = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$ is bounded. Why? We see that a possible upper bound is 1, which belongs to A_4 . What about a lower bound? The number 0 is a lower bound, however 0 does not belong to A_4 . Below we will see that there is no lower bound that belongs to A_4 .

5. The subset $A_5 = \{x \mid |x| > 100\}$ is unbounded.

Let us take a closer look at the lower bound of $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$:

Lemma 2.6: Any lower bound of the set $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$ does not belong to A_4 .

Proof. By contradiction, suppose that $\exists a \in A_4$ such that $a \leq x$ for all $x \in A_4$. Since $a \in A_4$, there exists $N \in \mathbb{N}$ such that $a = \frac{1}{N}$. Observe that $\frac{1}{N+1}$ is also an element of A_4 and $\frac{1}{N+1} < \frac{1}{N} = a$. But this contradicts our assumption that a is a lower bound for A_4 . Hence no element in A_4 can be a lower bound of A_4 . $\square$

In fact, we can prove a stronger statement:

Lemma 2.7: The set $A_4 = \{\frac{1}{n} \mid n \in \mathbb{N}_+\}$ does not have a positive lower bound.

Proof. By contradiction, assume that there exists $r > 0$ that is a lower bound for A_4 . Define $N = \lceil \frac{1}{r} \rceil$ to be the first integer greater than or equal to $\frac{1}{r}$. Then $N + 1 > \frac{1}{r}$ and consequently $\frac{1}{N+1} < r$. But $\frac{1}{N+1} \in A_4$, in contradiction to the assumption that r is a lower bound. Hence there is no positive lower bound, and 0 is the *greatest lower bound*. $\square$

Supremum and infimum

Let $A \subset \mathbb{R}$ be a subset.

- The **supremum** (if it exists) of A (also called the **least upper bound**, **l.u.b.**) is the smallest of all upper bounds of A . It is denoted

$$s = \sup A$$

and it fulfils the following two conditions:

1. $\forall x \in A, x \leq s$,
2. $\forall r < s \exists x \in A \text{ s.t. } x > r$.

If there is no such number, we define $\sup A = +\infty$.

- The **infimum** (if it exists) of A (also called the **greatest lower bound**, **g.l.b.**) is the largest of all lower bounds of A . It is denoted

$$\ell = \inf A$$

and it fulfils the following two conditions:

1. $\forall x \in A, x \geq \ell$,
2. $\forall r > \ell \exists x \in A \text{ s.t. } x < r$.

If there is no such number, we define $\inf A = -\infty$.

Note that in the above definition, we defined *the* supremum and *the* infimum, implying that these two numbers are unique. *A priori*, that is not obvious (even though it is true). It requires a proof.

Proposition 2.8: For any subset $A \subseteq \mathbb{R}$, there are unique elements $\ell, s \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$ such that $\ell = \inf A$ and $s = \sup A$.

Proof. The *existence* of these elements is guaranteed by the *Completeness Axiom*, stated immediately below. The *uniqueness* of these elements is left as an exercise. *Hint: prove it by contradiction.* $\square$

The Completeness Axiom

The property described above — that the real number line has “no gaps” — is in fact an *axiom*: it cannot be derived from the other properties of $\mathbb{R}$ (it is precisely the property that $\mathbb{R}$ has and $\mathbb{Q}$ does not). We state it here in the form we will use throughout these notes:

Completeness Axiom. Every nonempty subset $A \subseteq \mathbb{R}$ that is bounded above (resp. below) has a supremum $\sup A \in \mathbb{R}$ (resp. infimum $\inf A \in \mathbb{R}$).

The supremum and the infimum might not belong to the set!

It is very important to remember that for a subset $A \subseteq \mathbb{R}$, its supremum and its infimum might *not* belong to it. We have seen it with A_4 above: its infimum is 0, yet $0 \notin A_4$. For $A_2 = \mathbb{N}$, the supremum is $+\infty$, which isn't a number, and in particular isn't an element of A_2 .

Keeping in mind the preceding comment, in the case that the supremum and/or infimum *do* belong to the set we give them another name:

Maximum and minimum

Let $A \subset \mathbb{R}$ be a subset. If $\sup A \in A$ then we say that the supremum is *attained*, and it is called the **maximum** of A and denoted

$$\max A.$$

Similarly, if $\inf A \in A$ then we say that the infimum is *attained*, and it is called the **minimum** of A and denoted

$$\min A.$$

The cardinality of subsets of $\mathbb{R}$

The *cardinality* of a set A is a measure of its size. The cardinality of a set containing finitely many elements is simply the number of elements: the cardinality of $A = \{-11, 600, \sqrt{17}\}$ is 3. The cardinality of the set $B = \{\text{Giulia}, \text{Sam}, \text{Amelia}\}$ is also 3. The concept of cardinality becomes more delicate when dealing with sets containing infinitely many elements.

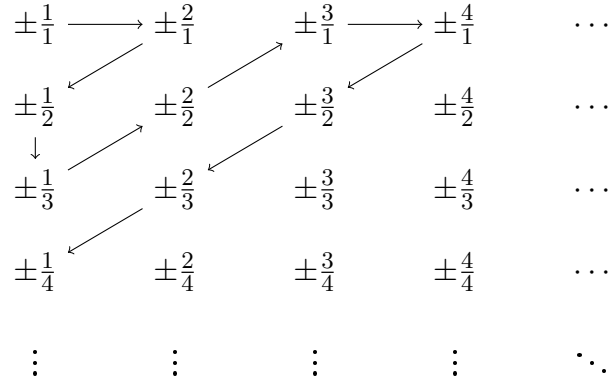
Countable sets

A subset $A \subseteq \mathbb{R}$ is said to be **countable** if it is possible to enumerate all its elements.

- Example 2.5:**
1. Any set with finitely many elements is countable.
 2. The set $\mathbb{N}_+ = \{1, 2, 3, \dots\}$ is trivially countable: we assign to the element $1 \in \mathbb{N}_+$ the number 1, to the element $2 \in \mathbb{N}_+$ the number 2, and so on.
 3. The sets $\mathbb{N}, \mathbb{Z}$ are also countable. The set of even natural numbers is also countable. *Can you prove it?*

Proposition 2.9: The set $\mathbb{Q}$ of rational numbers is countable.

Proof. To prove that a set is countable we need to demonstrate that we can enumerate its elements. All non-zero rational numbers are of the form $\pm \frac{p}{q}$ where $p, q \in \mathbb{N}_+$. These can be organized in a table as follows:



The first row contains all rationals with $q = 1$, the second row contains all rationals with $q = 2$, and so on. At the same time, the first column contains all rationals with $p = 1$, the second column contains all rationals with $p = 2$, and so on. Thus every non-zero rational appears within this table. In fact, every non-zero rational appears infinitely many times in this table, because for every $\pm \frac{p}{q}$, the table also contains $\pm \frac{2p}{2q}$, $\pm \frac{3p}{3q}$, and so on.

The arrows in the table demonstrate a strategy for enumerating the rationals. Since 0 doesn't appear in the table, we start by counting it: we assign to it the number 1. Then we enumerate the elements $\pm \frac{1}{1}$ to which the numbers 2 and 3 are assigned. Next we move on to $\pm \frac{2}{1}$ to which the numbers 4 and 5 are assigned. Then $\pm \frac{1}{2}$, to which 6 and 7 are assigned, $\pm \frac{1}{3}$ to which 8 and 9 are assigned, $\pm \frac{2}{2}$ to which 10 and 11 are assigned, and so on. Eventually to every rational will be assigned a natural number, completing the proof. $\square$

Theorem 2.10: The set $\mathbb{R}$ of real numbers is *not* countable.

Proof. The following proof, *by contradiction*, due to Georg Cantor, goes back to the late 19th century. It's enough to just look at the real numbers between 0 and 1. Each $x \in (0, 1)$ has a decimal representation:

$$x = 0. a_1 a_2 a_3 a_4 a_5 \dots \quad \text{where } a_i \in \{0, 1, \dots, 9\} \text{ for every } i \in \mathbb{N}_+.$$

Suppose, by contradiction, that the real numbers in $(0, 1)$ were countable. Then we can enumerate all $x \in (0, 1)$. Let's enumerate them as $r_1, r_2, \dots, r_n, \dots$. Each of these has a decimal representation: $r_n = 0. c_{n,1} c_{n,2} c_{n,3} c_{n,4} \dots$. Let's write the following table, with r_1 on the first line, r_2 on the second, and so on:

$r_1 =$	0.	$c_{1,1}$	$c_{1,2}$	$c_{1,3}$	$c_{1,4}$	$c_{1,5}$	$\dots$
$r_2 =$	0.	$c_{2,1}$	$c_{2,2}$	$c_{2,3}$	$c_{2,4}$	$c_{2,5}$	$\dots$
$r_3 =$	0.	$c_{3,1}$	$c_{3,2}$	$c_{3,3}$	$c_{3,4}$	$c_{3,5}$	$\dots$
$r_4 =$	0.	$c_{4,1}$	$c_{4,2}$	$c_{4,3}$	$c_{4,4}$	$c_{4,5}$	$\dots$
$r_5 =$	0.	$c_{5,1}$	$c_{5,2}$	$c_{5,3}$	$c_{5,4}$	$c_{5,5}$	$\dots$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Remember that each digit $c_{i,j}$ is an integer between 0 and 9. Let's now look at the numbers $c_{i,i}$ on the *diagonal*. Consider a real number between 0 and 1 defined as:

$$r = 0. d_1 d_2 d_3 d_4 d_5 \dots \quad \text{where } d_i \neq c_{i,i} \quad \forall i \in \mathbb{N}_+$$

(and also d_i shouldn't be 0 or 9: *can you think why?*). Then

$$\begin{array}{ll} r \neq r_1 & \text{since the } \textit{first} \text{ digit in their expansions is different} \\ r \neq r_2 & \text{since the } \textit{second} \text{ digit in their expansions is different} \\ r \neq r_3 & \text{since the } \textit{third} \text{ digit in their expansions is different} \\ \vdots & \\ r \neq r_n & \text{since the } \textit{nth} \text{ digit in their expansions is different} \\ \vdots & \end{array}$$

Hence r isn't equal to any of the reals that have been enumerated. However, that is a contradiction to the assumption that all reals in $(0, 1)$ have been enumerated. $\square$

This method of proof is called a **diagonal argument**, and has become an important technique for proving various results in the years since Cantor introduced it.

2.4 Cartesian product

Ordered pairs and Cartesian product

Let X and Y be two nonempty sets. Then we define an **ordered pair** to be

$$(x, y)$$

where $x \in X$ and $y \in Y$. The set of all ordered pairs from X and Y is called the **Cartesian product of X and Y** and is denoted $X \times Y$:

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}.$$

When $X = Y$ we often write X^2 rather than $X \times X$.

The order of elements in an ordered pair is important: the first *component* belongs to X and the second *component* belongs to Y . Thus, the ordered pair (x, y) is *fundamentally different* from the set $\{x, y\}$.

Example 2.6: 1. One of the most important examples of a Cartesian product, which we will often use, is that of the **plane** $\mathbb{R}^2$. This is the set of all ordered pairs of real numbers:

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

This is called the **Cartesian plane**.